

# Introduction to Security

Sam Ogden

# DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

# Lecture Objectives

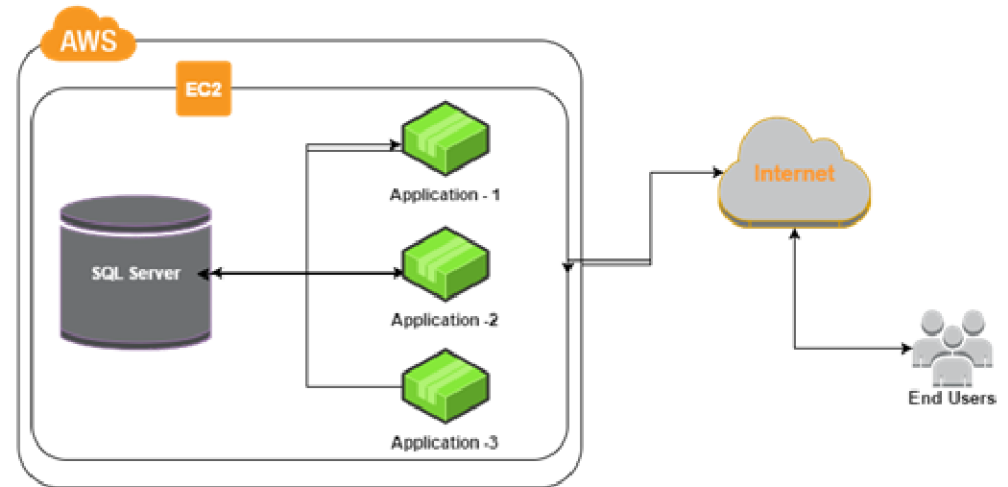
- After this lecture, you should be able to:
- Explain why security is an essential consideration for operating systems
- Discuss the goals of security
- Core security concepts

# What we've done so far

- So far we've virtualized resources:
  - Limited Direct Execution for processes
  - Paging for memory
- That keeps processes apart from each other
- But isolation alone does not answer how shared resources should be controlled
- Is that really enough?

# Resource Sharing is still important

- Not all applications can share memory with just threads
- Sometimes we need to communicate between processes
- Should all processes have equal footing?



# What might we want to protect?

- Security starts by naming the assets that matter
- Process memory
  - Passwords, personal information, execution pathways
- Files
  - User passwords are stored in a file. Who should be able to write?
- I/O devices
  - e.g. Who should be able to print?
  - e.g. Who should be able to send network messages?
- Resource decisions!

# **We need the OS to help us with this!**

**The OS already controls the resources we need to protect**

**That makes it the natural place to enforce security policy**

# OS Responsibilities

- The OS owns memory management and sharing, so it must protect each process's memory
- The OS starts, pauses, and terminates processes, so it must make reasonable choices
- Resources are made available by the OS, so it must allocate them safely



# Security Goals

- **Confidentiality:** keep information secret
- **Integrity:** keep information correct
- **Availability:** keep the system up and running

# Example: A cryptominer virus

- Mines crypto using your GPU, without your permission
  - Increases your power bill
  - Reduces your game's FPS

# Example: A cryptominer virus

- Mines crypto using your GPU, without your permission
  - Increases your power bill
  - Reduces your game's FPS

***What goals does this circumvent?***

# Example: A cryptominer virus

- Mines crypto using your GPU, without your permission
  - Increases your power bill
  - Reduces your game's FPS

***What goals does this circumvent?***

*Availability*

# Security design is about tradeoffs

- The goal is not just to block attacks
- The goal is to protect the system without making it unusable
- Good security design reduces damage when something goes wrong

# Design Principles

## Keep it simple

- Economy of mechanism
- Open design

## Limit blast radius

- Least privilege
- Least common mechanism
- Acceptability

## Enforce policy carefully

- Fail-safe defaults
- Complete mediation
- Separation of privilege

# Encouraging good security

- Weakest links will break first
  - Breaking a cryptographic hash is really hard
  - Breaking “password” (i.e. the literal word as a password) is really easy





# Encouraging good security

- Weakest links will break first
  - Breaking a cryptographic hash is really hard
  - Breaking “password” (i.e. the literal word as a password) is really easy

## A CRYPTO NERD'S IMAGINATION:

HIS LAPTOP'S ENCRYPTED.  
LET'S BUILD A MILLION-DOLLAR  
CLUSTER TO CRACK IT.

NO GOOD! IT'S  
4096-BIT RSA!

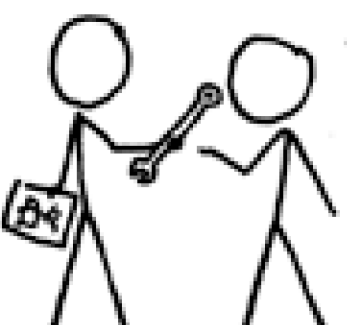
BLAST! OUR  
EVIL PLAN  
IS FOILED!



## WHAT WOULD ACTUALLY HAPPEN:

HIS LAPTOP'S ENCRYPTED.  
DRUG HIM AND HIT HIM WITH  
THIS \$5 WRENCH UNTIL  
HE TELLS US THE PASSWORD.

GOT IT.



# System Calls - Not all are equal

- `exit()`
- `getpid()`
- `getppid()`
- `time()`
- `wait()`
- `fork()`
- `mmap()`
- `mprotect()`
- `socket()`
- `recv()`

# Summary

- Security is essential to any system
- Virtualization and sharing are useful, but they create policy decisions that the OS must enforce
- The security goals are confidentiality, integrity, and availability
- Good security comes from simple mechanisms, careful checks, and designs people can actually use
- In practice: focus on what works, and make sure it still works when things go wrong